

Quality-Stratified Multimodal Fusion for Skin Lesion Malignancy Classification on MCR-SL

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Outline

- 1 Motivation
- 2 Dataset
- 3 Our Contributions
- 4 Architecture
- 5 Training & Evaluation
- 6 Core Results
- 7 Quality-Adaptive Loss
- 8 Robustness Analyses
- 9 Validation Discipline
- 10 Comparison & Conclusion

Why This Problem?

- Deep learning has reached dermatologist-level accuracy on **image-only** skin lesion classification (Esteva et al., 2017; HAM10000).
- But a real dermatologist never diagnoses from an image alone — age, sex, lesion site, referral history all matter.
- A smaller line of work fuses images with **patient metadata** (e.g. MetaBlock on PAD-UFES-20) and shows it helps.
- **Missing piece:** none of these datasets record how *good* the diagnostic image itself was.
 - Real clinical images vary hugely in quality (lighting, focus, hair, blur).
 - No prior benchmark could ask: *do errors concentrate on poor-quality images?*

The Opportunity: MCR-SL

A brand-new dataset (Castro-Fernandez et al., *Data*, 2025) closes this gap:

- Every diagnostic image has an expert-assigned **1–10 quality rating**.
- Rich structured metadata per lesion/subject (22 fields).
- Partial histopathology confirmation (ground-truth beyond panel opinion).

Key fact

No prior published benchmark exists on MCR-SL. We are first.

This lets us ask a question no earlier skin-lesion benchmark could:

Does model performance degrade on lower-quality images, and can we train the model to be more robust to that?

MCR-SL Dataset at a Glance

Property	Value
Subjects	60
Lesions (total / usable for binary task)	240 / 234
Malignant / non-malignant / undetermined	42 / 192 / 6
Clinical images (usable)	779
Dermoscopic images (usable)	1352
Histopathology-confirmed lesions	28
Metadata fields used (categorical / numeric)	16 / 4
Lesions with full expert quality coverage	238
Unified diagnosis classes	9

“Usable” = has a lesion record match and an existing image file on disk.

Task Definition

Primary task: Binary classification

Malignant vs. non-malignant, per lesion.

Ground truth: `unified_diagnosis` field — histopathology-confirmed where available, expert-panel consensus otherwise.

Secondary / exploratory task: 9-class diagnosis

NEV, SK, BCC, AK, ATY, MEL, SCC, ANG, DF.

Caveat: several classes have <10 lesions (MEL=8, SCC=4, ANG=4, DF=2) — reported as exploratory only, never as robust per-class statistics.

What This Work Does

- ① First published benchmark on MCR-SL: image-only vs. late-fusion vs. **channel-gated fusion**, under a subject-disjoint 5-fold protocol.
- ② **A quality-adaptive loss reweighting scheme** (`hard_mining`) that uses MCR-SL's expert quality ratings to weight the training loss — **and a shuffled-quality control that validates the gain is genuinely quality-driven**, not generic reweighting.
- ③ **Six robustness analyses** enabled by MCR-SL's quality ratings, expert certainty, histopathology subset, and multi-lesion subjects.
- ④ **Three apparent positive findings that we overturned ourselves** using proper controls — a validation discipline we report as part of the contribution, not an appendix.
- ⑤ Final result: **0.840 ± 0.095 balanced accuracy** (`hard_mining` + TTA + multi-image averaging), *within noise of* the closest verified binary comparator (0.836) using $\sim 10\times$ less training data.

The architecture is unchanged from standard components. The contribution is the *loss weighting* and the *validation*, not a new network.

Why MCR-SL, Not a Bigger Dataset?

The honest answer: it is not about scale. It is about a field that only exists here.

Dataset	Size	Expert per-image quality rating?
ISIC (2019/2020)	25,000–35,000+	No
HAM10000	10,015	No
PAD-UFES-20	2,298	No
MCR-SL	240 lesions	Yes — 1–10, per diagnostic image

- Larger datasets buy **statistical power**. None of them can answer this project's research question, because **the field the mechanism depends on does not exist in them**.
- MCR-SL's ratings are **real expert judgements of the photograph**, tied to the specific image used for each diagnosis — not an automated blur score or a learned proxy.
- We accept the cost of small N (wide error bars, reported throughout) in exchange for asking a question that is **unanswerable elsewhere**.

Has Our Research Question Been Answered?

The project asks **two separable questions**. They get two different answers.

Q1: Does performance degrade on lower-quality images? — **No (null result)**

Spearman(rating, error) = -0.093 , $p = 0.158$, $N = 231$; tercile pattern non-monotonic (0.776 / 0.914 / 0.868). Reported as a **genuine null** — neither confirmed nor ruled out. **Not hidden**.

Q2: Can the model be trained to be more robust to quality? — **Yes, with evidence**

hard_mining loss reweighting: **+0.037** balanced accuracy, **+0.092** sensitivity, **+0.025** AUROC.

Validated, not asserted: a shuffled-quality control (same weight distribution, randomised lesion→rating mapping) **collapses the entire gain**.

Q1 being null does *not* invalidate Q2 — a model can be made more robust to a factor that shows no significant baseline correlation at this N .

Architecture — Two Encoders

Image Encoder

- EfficientNet-B0, ImageNet-pretrained
- Input: dermoscopic image, 224×224
- Output: 1280-d pooled vector *and* the last conv feature map ($1280 \times 7 \times 7$)

Metadata Encoder

- 16 categorical fields $\rightarrow$ learned embeddings (with explicit “unknown” category — *no imputation*)
- 4 numeric fields (age, height, weight, diameter) $\rightarrow$ z-scored using *train-fold-only* statistics + missingness bit
- Concat $\rightarrow$ 2-layer MLP $\rightarrow$ 128-d metadata vector

Both encoders feed into one of two fusion strategies (next slide).

Architecture — Two Fusion Strategies

1. Late fusion (baseline)

concat(pooled image vector, metadata vector) $\rightarrow$ Linear+ReLU+Dropout (256-d) $\rightarrow$ classifier.
This is the standard MetaBlock-style approach from prior literature.

2. Channel-gated fusion (our main method)

metadata vector $\rightarrow$ Linear $\rightarrow$ Sigmoid $\rightarrow$ a 1280-d **gate** $\in [0, 1]^{1280}$
 $\rightarrow$ multiply elementwise against the image's conv feature map, channel-by-channel
 $\rightarrow$ global pool $\rightarrow$ same 256-d projection $\rightarrow$ classifier.

Idea: let patient metadata decide **which visual channels matter**, instead of just concatenating two vectors. Same mechanism as Squeeze-and-Excitation, but the gating signal comes from metadata, not from the image itself.

Architecture — Three Prediction Heads

All three heads sit on top of the **same** 256-d fused vector; only their outputs and losses differ.

1. Binary head (main task, always on)

256-d $\rightarrow$ Linear $\rightarrow$ 1 logit $\rightarrow$ sigmoid $\rightarrow P(\text{malignant})$.

Loss: weighted BCE (or focal loss), class weights recomputed per training fold ($\sim 1:4.6$ malignant:non-malignant).

This is the only head whose output is reported as the paper's headline results.

2. Auxiliary 9-class head (exploratory)

256-d $\rightarrow$ Linear $\rightarrow$ 9 logits $\rightarrow$ softmax $\rightarrow$ unified diagnosis (NEV, SK, BCC, ...).

Loss: class-weighted CE, downweighted to $\sim 0.3\text{--}0.5\times$ the binary loss.

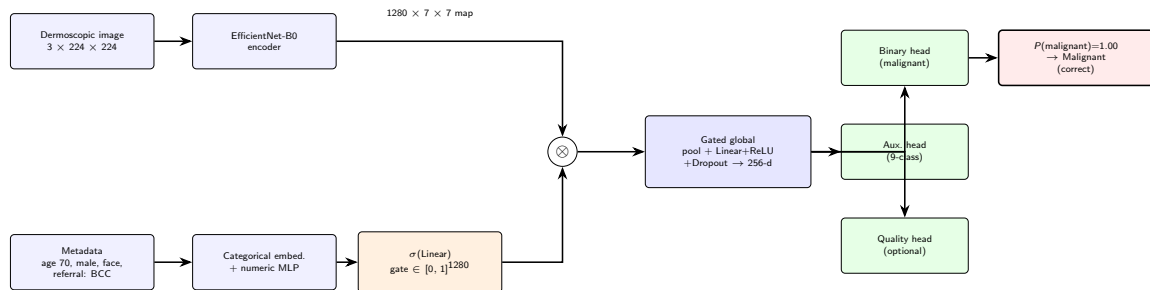
Reported only as an exploratory table — several classes have <10 lesions.

3. Quality-regression head (optional, quality-aware variant only)

256-d $\rightarrow$ Linear $\rightarrow$ 1 scalar $\rightarrow$ predicted mean expert quality rating (1–10). Loss: MSE, weighted $0.1\text{--}0.2\times$.

*Tests whether **predicting** quality improves robustness — it did not. Later superseded by **weighting the loss with quality instead**, which did work.*

Full Pipeline (with a Real Example)



Example: lesion L0189, histopathology-confirmed basal cell carcinoma — one of the model's most confident correct predictions.

Training / Evaluation Protocol

Subject-disjoint stratified 5-fold cross-validation

- Split by `subject_id`, never by lesion — a subject can have multiple lesions, and images of the same lesion are near-duplicates.
- Folds balanced by greedily equalizing malignant-lesion counts per fold.
- For each rotation: 1 fold held out for **test**, 1 disjoint fold held out only for **checkpoint selection** (val), rest for training.
- **No hyperparameter tuning against final test-fold results.**

Losses

- Binary head: weighted BCE, class weights recomputed per training fold ($\sim 1:4.6$ malignant:non-malignant overall).
- Auxiliary 9-class head: class-weighted CE, downweighted ($\sim 0.3\text{--}0.5\times$), exploratory only.
- (Optional) Quality-regression head: low weight ($0.1\text{--}0.2$), predicts mean expert rating.

The 5 Folds, Visualized

60 subjects $\rightarrow$ 5 disjoint folds (balanced by malignant-lesion count). Every rotation uses a **different** test fold and a different, disjoint val fold — so all 5 folds eventually serve as the test fold exactly once.

Rotation	Fold 0	Fold 1	Fold 2	Fold 3	Fold 4
1	Test	Val	Train	Train	Train
2	Train	Test	Val	Train	Train
3	Train	Train	Test	Val	Train
4	Train	Train	Train	Test	Val
5	Val	Train	Train	Train	Test

- **Train** folds: model weights are updated here.
- **Val** fold: used only to pick the best training checkpoint (never influences weights).
- **Test** fold: touched exactly once, only for final reported metrics — never used for any decision.

Final numbers = mean $\pm$ std of the 5 **test**-fold results (one per rotation).

Evaluation Metrics — What Each One Means

Metric	Definition	Why it matters here
Accuracy	(correct predictions) / (all predictions)	Can be misleading: predicting “non-malignant” always still scores $\sim 82\%$
Balanced accuracy	average of sensitivity and specificity	Primary metric — corrects for the malignant class being a minority ($\sim 18\%$)
Macro-F1	F1 (precision/recall harmonic mean) averaged equally across both classes	Rewards doing well on <i>both</i> classes, not just the majority one
Sensitivity (recall)	(malignant caught) / (all actual malignant)	Missed malignancy = missed cancer diagnosis — the costliest error clinically
Specificity	(non-malignant cleared) / (all actual non-malignant)	Low specificity means unnecessary biopsies/patient anxiety
AUROC	probability the model ranks a random malignant lesion above a random non-malignant one	Threshold-independent measure of overall ranking quality

All reported as mean $\pm$ std across the 5 test folds, plus a confusion matrix aggregated across all folds.

Core Ablation: Image-Only vs. Late Fusion vs. Channel-Gated

Metric	Image-only	Late fusion	Channel-gated
Accuracy	0.813 ± 0.020	0.831 ± 0.052	0.833 ± 0.070
Balanced acc.	0.784 ± 0.073	0.768 ± 0.090	0.783 ± 0.085
Macro F1	0.733 ± 0.038	0.743 ± 0.051	0.757 ± 0.073
Sensitivity	0.697 ± 0.187	0.644 ± 0.191	0.672 ± 0.159
Specificity	0.872 ± 0.061	0.891 ± 0.021	0.895 ± 0.048
AUROC	0.895 ± 0.053	0.851 ± 0.080	0.881 ± 0.065

Channel-gated wins macro-F1, but does **not** dominate every metric.

Image-only actually has higher sensitivity and AUROC — reported honestly, not hidden.

Balanced accuracy is effectively tied between image-only and channel-gated (0.784 vs. 0.783).

Quality-Aware Variant — A Negative Result

Added an auxiliary head to predict image quality, hoping it would make the model more robust.

Metric	Channel-gated	+ Quality-aware
Balanced accuracy	0.783	0.740
Sensitivity	0.672	0.578
Specificity	0.895	0.902
AUROC	0.881	0.879

It **underperformed** on 5 of 6 metrics — became more conservative (small specificity gain, big sensitivity loss). Worse direction for a cancer-screening task. Reported honestly as a negative result.

Follow-Up Intervention Sweep (11 configurations)

Configuration	Acc.	Bal. acc.	Sens.	Spec.	AUROC
Focal loss	0.858	0.809	0.694	0.923	0.877
Dermoscopy preprocessing	0.828	0.793	0.736	0.851	0.875
Contrastive auxiliary loss	0.812	0.770	0.669	0.872	0.873
Alt. optimizer (AdamW+cosine)	0.822	0.757	0.661	0.853	0.845
Test-time augmentation (TTA)	0.841	0.788	0.672	0.904	0.897
Multi-image averaging	0.854	0.813	0.719	0.907	0.906
Sharpness-aware min. (SAM)	0.858	0.810	0.694	0.927	0.904
SAM + multi-image	0.851	0.800	0.672	0.928	0.917
SAM + TTA	0.861	0.822	0.719	0.925	0.908
SAM + TTA + multi-image	0.856	0.810	0.694	0.925	0.915
Preprocess.+focal+SAM+TTA	0.818	0.759	0.622	0.895	0.893

Best in *this* sweep: SAM + TTA $\rightarrow$ 0.822. Stacking everything together **hurt** rather than helped. **All of these were later superseded by the quality-adaptive loss (next section).**

Quality-Adaptive Loss: The Mechanism

Every lesion has a mean expert quality rating $q \in [1, 10]$. We use it to **weight the training loss per sample** — not to predict it as an output (that was the auxiliary head, which failed).

trust — down-weight poor images

$$w = 0.5 + \frac{q-1}{9} \Rightarrow [1, 10] \rightarrow [0.5, 1.5]$$

“Low-quality images are less reliable, so trust them less.”

hard_mining — up-weight poor images

$$w = 1.5 - \frac{q-1}{9} \Rightarrow [1, 10] \rightarrow [1.5, 0.5]$$

“Force the model to work harder on the difficult images.”

$$\mathcal{L}_i = \underbrace{w_i^{\text{class}}}_{\text{existing class weight}} \times \underbrace{w_i^{\text{quality}}}_{\text{new}} \times \text{BCE}(\hat{y}_i, y_i)$$

Lesions with no valid rating get a neutral weight of 1.0. Fixed range, two directions, **no grid search** — the range was not tuned. Architecture, optimiser, folds and checkpoint rule are **identical** to the baseline; only the loss weighting changes.

Validating the Loss: The Shuffled-Quality Control

The obvious objection

“Maybe *any* per-sample reweighting acts as a regulariser. Perhaps the quality ratings are irrelevant and you have just found noise that happens to help.”

The control: within each training fold, **shuffle** the quality ratings across lesions. Identical weight *distribution*, but the lesion→rating correspondence is destroyed. Test folds always use **real, unshuffled**

ratings. Everything else identical.

Mechanism	Bal. acc.	Sens.	High–Low gap
Plain baseline (no quality)	0.783	0.672	0.091
hard_mining (real ratings)	0.820	0.764	0.091
hard_mining (shuffled)	0.775	0.667	0.016
trust (real ratings)	0.788	0.725	0.082
trust (shuffled)	0.763	0.672	0.021

- hard_mining **passes**: shuffling collapses it to 0.775 / 0.667 — *back to baseline*. The gain **requires the real quality signal**.
- trust **fails**: its apparently-narrower tercile gap (0.082) is *beaten* by the shuffled control (0.021). Revealed as a reweighting artifact — reported as a negative result.

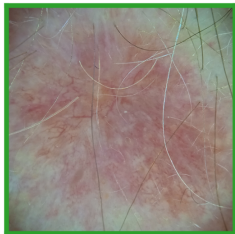
Where Does the Gain Actually Come From?

Decomposing the final result — how much is genuinely about quality?

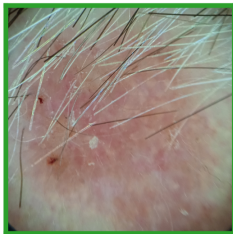
Step	Configuration	Uses quality?	Bal. acc.	Δ vs base	Sens.
0	Baseline (channel-gated)	No	0.783	—	0.672
1	+ hard_mining loss	Yes	0.820	+0.037	0.764
2	+ TTA (eval-time)	No	0.833	+0.050	0.808
3	+ multi-image (eval-time)	No	0.840	+0.057	0.783
ctrl	hard_mining, shuffled	Fake	0.775	-0.008	0.667

- Of the total **+0.057** gain, **+0.037** ($\approx 65\%$) is the validated, quality-specific contribution.
- The remaining $\approx 35\%$ is generic eval-time technique (TTA, multi-image averaging) with **nothing to do with quality** — stated explicitly so the novel part is not oversold.
- Steps 2–3 require **no retraining** at all.

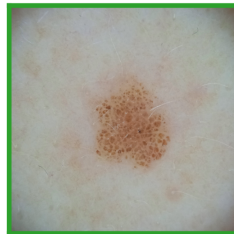
Example Predictions (Best Method)



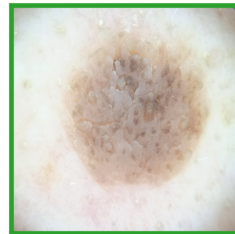
Age 70 / Male / Face
Referral: BCC
Ground truth: Malignant
Predicted: Malignant, $P=1.00$



Age 57 / Male / Face
Referral: BCC
Ground truth: Malignant
Predicted: Malignant, $P=1.00$



Age 50 / Female / Back
Referral: Voluntary sample
Ground truth: Non-malignant
Predicted: Non-malignant, $P=0.00$



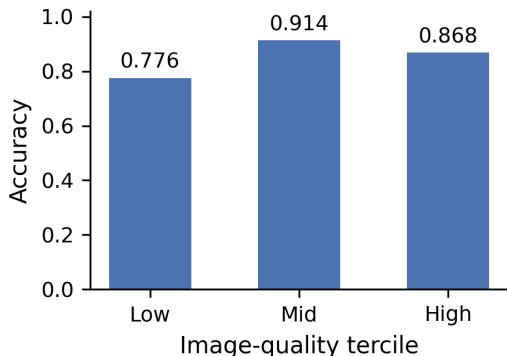
Age 64 / Male / Back
Referral: SK
Ground truth: Non-malignant
Predicted: Non-malignant, $P=0.00$

Highest-confidence *correct* predictions — illustrative only, not representative of overall performance (see full metrics above).

Why Robustness Analyses Are the Real Contribution

- Beating a prior number was never really possible — **no prior benchmark exists on MCR-SL.**
- What MCR-SL uniquely offers: per-image expert quality ratings + partial histopathology confirmation.
- So the paper's actual novelty is asking questions no earlier skin-lesion benchmark could ask:
 - ① Does performance drop on lower-quality images?
 - ② Can we train the model to be more robust to quality variation?
 - ③ Does performance differ on histopathology-confirmed vs. panel-consensus lesions?
 - ④ Does the model “look at” the same metadata fields that the dataset's own statistics flagged as significant?

1. Quality-Stratified Performance



Tercile	N	Acc.	Sens.
Low	85	0.776	0.667
Mid	93	0.914	0.733
High	53	0.868	0.500

Pattern is **not monotonic**. Spearman correlation (rating vs. error) = -0.093 , $p = 0.158$ — not significant at $N = 231$. No strong quality-performance relationship detected (neither confirmed nor ruled out).

Robustness Across the Quality Ranking

Does the validated loss hold up **at every quality level**, not just on average?

Quality tercile	N	Accuracy		Sensitivity	
		plain	hard_mining	plain	hard_mining
Low	85	0.776	0.776	0.667	0.810
Mid	93	0.914	0.903	0.733	0.733
High	53	0.868	0.868	0.500	0.667
High—Low accuracy gap		0.091	0.091		

- **The accuracy gap is unchanged** ($0.091 \rightarrow 0.091$). `hard_mining` does *not* flatten the quality-tercile profile — stated plainly.
- **But sensitivity rises where it matters most:** on the *lowest*-quality tercile, malignant recall goes $0.667 \rightarrow \mathbf{0.810}$. Under `hard_mining`, sensitivity is now **highest on the worst images** — exactly the on-mechanism behaviour up-weighting them should produce.

Honest framing: the *loss* improves quality robustness; the *architecture* is unchanged and we make no claim it is inherently quality-robust. Malignant counts within each tercile are small — directional evidence, not a decisive test.

2. Quality-Aware Training — Did It Help?

Tercile	N	Acc. (plain)	Acc. (quality-aware)
Low	85	0.776	0.729
Mid	93	0.914	0.903
High	53	0.868	0.830

- High—Low accuracy gap: **0.091 without** quality-awareness vs. **0.101 with** it.
- The intervention **widened**, rather than narrowed, the quality-performance gap.
- Direct negative result — reported as such.

3. Histopathology-Confirmed vs. Panel-Consensus

Group	N	Accuracy
Histopathology-confirmed	28	0.750
Panel-consensus-only	203	0.867

- Model performs *worse* on the histopathology-confirmed subset.
- Plausible reason: histopathology is ordered more often for clinically *ambiguous or suspicious* lesions in real practice — these are inherently harder cases.
- $N = 28$ is too small for a confidence interval — treated as a **qualitative** finding only.

4. Metadata Importance — Does the Model “Look at” the Right Fields?

Method: ablate each field (replace with “unknown”), measure change in output logit.

Highest ranked	Importance
Organ transplant history	0.046
Lesion status when captured	0.041
Referral diagnosis*	0.024
Sex*	0.015
Location group*	0.011
Lowest ranked	
Age / Weight / Diameter* / Height	≤ 0.0011

* = field the dataset paper found significantly associated with malignancy.

Referral diagnosis, sex, location group align with the dataset paper. But **diameter** ranks lowest — likely an architectural artifact (12-d categorical embeddings vs. 1 scalar), not true unimportance.

Three Times We Almost Reported a False Positive

Each of these **looked like a result**. A control overturned it. We report all three.

What it looked like	The check	What it actually was
trust narrowed the quality-tercile gap $0.091 \rightarrow 0.082$ — a second quality-robustness win	Shuffled-quality control	Reversed. The <i>shuffled</i> control produced an even narrower gap (0.021). A generic reweighting artifact, not a quality effect.
Expert diagnostic certainty predicted model error, $\rho = -0.175$, $p = 0.008$ — apparently significant	Stratify by malignant / non-malignant class	Reversed. Dissolves within both strata ($p = 0.083$ / $p = 0.559$). Malignant lesions cluster at low certainty, so the pooled correlation was class-mix confounding .
“Train on all images per lesion, not just one” — a large untapped data opportunity	Verify the loader before building anything	False premise. Image-level training had been the default since run one ($5.65\times$ more samples than lesions). Caught before any effort was wasted.

At $N = 234$ with $\sim 8\text{--}9$ malignant lesions per fold, **an uncontrolled first look reliably produces plausible positives**. We treat catching them as part of the contribution — and it is the most transferable lesson here.

What Is Our Novelty, Precisely?

Not the architecture. Not a new loss function class. Not state of the art.

- 1 **First use of real expert-assigned quality labels for quality-adaptive loss reweighting in medical image classification.** The closest precedents — **AdaFace** (CVPR 2022) and **MagFace** — come from face recognition and must use *feature-norm as a proxy*, because face datasets carry no expert quality annotation. MCR-SL does.
- 2 **The opposite weighting direction works here.** AdaFace/MagFace *de-emphasise* low-quality samples; our `hard_mining` *up-weights* them, and that is the variant that wins.
Plausible mechanism: a dermatologist can often still diagnose correctly from a poor photograph, so the label remains valid and the hard case is worth learning from. In face ID, poor quality can make identity genuinely **unrecoverable** — so de-emphasis is right there, and wrong here.
- 3 **Distinguished from the nearest skin-lesion prior work.** arXiv:2308.02119 (HAM10000) performs expert-informed sample reweighting, but weights by **how the label was confirmed** (diagnostic method). We weight by a **direct expert rating of the photograph itself** — a different signal answering a different question.
- 4 **The validation discipline is part of the claim.** The shuffled-quality control, and the three reversed findings, are reported as contributions — not buried.

Comparison with the Literature

Source	Task	N	Bal. acc.
This work (channel-gated baseline)	binary	234 lesions	0.783 ± 0.076
This work (+ hard mining)	binary	234 lesions	0.820 ± 0.078
This work (+ TTA + multi-image)	binary	234 lesions	0.840 ± 0.095
DiffMIC, PAD-UFES-20 (2025)	binary	2298 images	0.836

This is NOT a claim of beating the state of the art

- Different dataset, different institution, no shared test set.
- Our fold-to-fold std (± 0.078 – 0.095) is **over $20\times$ larger** than the 0.004 nominal gap — **statistically indistinguishable**.
- The last $+0.020$ came from *eval-time averaging*, not a better method.

Correct framing: **within noise of the closest verified binary comparator, using $\sim 10\times$ less training data.**

Encouraging context for a 240-lesion dataset — not a win.

Cross-Dataset Check: DeepDRiD (Not Completed)

What we set out to test

Does `hard_mining` transfer to a **second medical imaging domain** with a differently-structured quality annotation? **DeepDRiD** (diabetic retinopathy, ISBI 2020) ships per-image quality annotations alongside DR severity grades — the closest analogue to MCR-SL's ratings outside dermatology.

Status: **gate not passed — no results, and none estimated**

A deliberate go/no-go gate ran **before** any training code was written. It did not pass in its time box: sustained throughput to the data host measured **26–64 KB/s** single-stream; required tooling (`git-lfs`) unavailable, and the archive endpoint ignores HTTP range requests, so the transfer cannot be parallelised.

No DeepDRiD number is reported, estimated, or implied anywhere in this deck.

Concrete future work. Having a gate that *can* fail — and honouring it — is the same discipline as the three reversed findings.

Limitations

- **Effect size vs. uncertainty:** our fold-to-fold std (± 0.078 – 0.095 balanced accuracy) is **larger than most differences we report**. The `hard_mining` effect ($+0.037$) is supported by the shuffled control, but **no two configs within ~ 0.05 should be called a win** over one another.
- **Single seed**, single institution, 234 usable lesions, ~ 8 – 9 malignant lesions per test fold — one lesion flipping moves a fold's sensitivity by ~ 0.11 .
- **Folds are subject-imbalanced** (5 / 6 / 16 / 16 / 16 subjects) — balanced on malignant-lesion count, not on subject count.
- **No external validation.** The DiffMIC comparison is cross-dataset context, not a benchmark. The DeepDRiD cross-domain check **was not completed**.
- Ground truth mostly panel-consensus (28/240 histopathology-confirmed); several 9-class categories have < 10 lesions — exploratory only.
- Quality terciles are modest ($N \approx 53$ – 93); **six analyses were run with no multiple-comparison correction**.
- Metadata-importance ranking is confounded by embedding-dimension imbalance (12-d categorical vs. 1 scalar numeric).
- A known `pos_weight` miscalibration ($\sim 16\%$, lesion-level weight applied to image-level batches) is **quantified and disclosed**, not silently corrected.

Conclusion & Future Work

What we found

- **First benchmark on MCR-SL.** Final: 0.840 ± 0.095 balanced accuracy — *within noise* of the closest verified binary comparator, with $\sim 10\times$ less data.
- **Q1 (does quality degrade performance?): null** — no significant relationship at $N = 231$.
- **Q2 (can we train for quality robustness?): yes** — `hard_mining` gives $+0.037$ balanced accuracy / $+0.092$ sensitivity, and a **shuffled control proves the gain depends on the real ratings**. Sensitivity on the *worst* images rises $0.667 \rightarrow 0.810$.
- **Three apparent positives we overturned ourselves**, plus quality-aware training, contrastive loss and alternative optimisers — all reported as negatives.

Future work

Complete the **DeepDRiD** cross-domain check; **multi-seed averaging** to shrink error bars that currently exceed most effect sizes; correct the disclosed `pos_weight` miscalibration; use the 779 unused clinical images with a modality indicator.

Thank you

Questions?